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## SYSTEMS AND METHODS FOR AUTOMATIC TERMINATION OF FLOW DUE TO NEEDLE DISLODGEMENT


### Abstract

Systems and methods for automatic flow termination for fluid delivery, including a housing configured for coupling a fluid delivery tube to a needle configured for subcutaneous delivery of fluid within a tissue of a patient and a spring-loaded activation mechanism having a first orientation corresponding to a condition where the housing is disposed substantially adjacent to the tissue and the needle lodged within the tissue and a second orientation corresponding to a condition where the housing is disposed away from the tissue or the needle being dislodged from the tissue. A flow termination mechanism is coupled to the activation mechanism and having an open configuration allowing flow from the fluid delivery tube to the needle when the activation mechanism is in the first orientation and a closed configuration substantially terminating flow from the fluid delivery tube to the needle when the activation mechanism is in the second orientation.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/233,270 filed Apr. 16, 2021, which is a continuation of U.S. application Ser. No. 16/244,445 filed Jan. 10, 2019 (now U.S. Pat. No. 10,994,075), which is a divisional of U.S. application Ser. No. 15/286,274 filed Oct. 5, 2016 (now U.S. Pat. No. 10,213,548), which is a continuation-in-part of International Application No. PCT/US2014/072573 filed Dec. 29, 2014, which claims the benefit of priority to U.S. Provisional Application No. 61/978,671 filed Apr. 11, 2014, each of which is herein incorporated by reference in its entirety for all purposes.

BACKGROUND
1. Technical Field
[0002] This disclosure pertains generally to vascular connections, and more particularly to detection and interruption of dislodged vascular connections.
2. Background Discussion
[0003] There are a number of techniques that provide a means by which to detect an errant flow of fluid through a vascular connection leading fluid from the outside of the body to the inside of the body. Common to many of these is the use of a ‘continuity sensor’ that looks for an interruption of energy-based signal or some mechanical connection from the tubing to the body. Such systems often use mechanical connectors, a small electrical current, a capacitance, a magnet or even ultrasound as a means of monitoring the fidelity of the connection between the body and the fluid passing element. Others use techniques designed to look for ‘wetness’ on the theory that a dislodged needle will leak fluid and fluid detection can be used as a surrogate marker for needle dislodgement.
BRIEF SUMMARY
[0004] An aspect of the present disclosure is a needle safety system or add-on to existing needles/tubing that uses a contact sensing mechanism on the patient's skin to determine when a given needle/tubing set that has been inserted into a patient has potentially become dislodged from that patient. This can occur when the tape holding a vascular access needle in place fails or the line is pulled out etc.
[0005] The system of the present disclosure offers important protection through the use of a fluid stop valve within the device that automatically deploys to stop the flow of fluid through a needle/tube when and only when, the needle delivering that fluid into the body is accidentally dislodged from the patient during fluid delivery. In hemodialysis that fluid is blood. In other cases, that fluid may be saline or medications. Vascular access is routinely performed in hospitals, clinics and other medical locations as well as the home (during home hemodialysis for example).

[0006] Another aspect is a device with a pinch valve configured in such a way that the valve is only activated by a mechanical linkage to a mechanical ‘skinsensing’ element in a needle system that has been pre-manufactured to include a compressible segment of tubing.
[0007] Another aspect is a system for sensing skin contact using a buttonlike sensor that comes straight out of the bottom of a needle body and halting flow using a blockage technique that involves rotating or sliding an opening from close to open within the needle valve.
[0008] The device of the current disclosure uses no external power, thus requires no batteries or cables, improving its ability to be adopted in medical workspaces that are complex and require simplified solutions. The device of the current disclosure is completely sterilizable and can be completely disposable. It can be manufactured relatively inexpensively using high-volume injection molding processes. It does not require extensive clinical training.
[0009] Further aspects of the technology will be brought out in the following portions of the specification, wherein the detailed description is for the purpose of fully disclosing embodiments of the technology without placing limitations thereon

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## Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS
[0010] The technology described herein will be more fully understood by reference to the following drawings which are for illustrative purposes only:
[0011] FIG. **1**A is a perspective view of a sensing mechanism employing a spring arm and pinching mechanism in accordance with the present disclosure.
[0012] FIG. **1**B is a perspective view of the sensing mechanism of FIG. **1**A in a released orientation.
[0013] FIG. **1**A′ is a perspective view of a sensing mechanism employing a spring arm and pinching mechanism in accordance with the present disclosure.
[0014] FIG. **1**B′ is a perspective view of the sensing mechanism of FIG. **1**A′ in a released orientation.
[0015] FIG. **2**A is a cross-sectional view of the sensing mechanism of FIG. **1**A.
[0016] FIG. **2**B is a cross-sectional view of the sensing mechanism of FIG. **1**A in a released orientation.
[0017] FIG. **3**A is a perspective view of the sensing mechanism of FIG. **1**A with the top cover removed.
[0018] FIG. **3**B is a perspective view of an alternative sensing mechanism of FIG. **1**A with integrated tube and housing.
[0019] FIG. **4** is an exploded perspective view of the sensing mechanism of FIG. **1**A.
[0020] FIG. **5**A shows a perspective view of a rotary-valve sensing mechanism in accordance with the present disclosure.
[0021] FIG. **5**B shows a cutout side view of the rotary-valve sensing mechanism of FIG. **5**A in an open configuration.
[0022] FIG. **5**C shows a cutout side view of the rotary-valve sensing mechanism of FIG. **5**A in a closed configuration.
[0023] FIG. **6**A through FIG. **6**C show sectional side views of a slider valve-based sensing system in accordance with the present disclosure.
[0024] FIG. **7**A shows an exploded perspective view of a preferred embodiment incorporating a rotary valve mechanism in accordance with the present disclosure.
[0025] FIG. **7**B is a side view of the rotary valve mechanism of FIG. **7**A in a closed configuration.
[0026] FIG. **7**C is a side view of the sealing mechanism of FIG. **7**A.
DETAILED DESCRIPTION

[0027] The sensing systems/mechanisms of the present disclosure are configured to detect separation of a vascular access needle or the associated fluid delivery tubing from the patient's skin, and act to terminate or restrict flow of the delivery fluid to the needle upon a detection or sensing of a needle separation/dislodgement.
[0028] For purposes of this disclosure, the term “sensing,” particularly with respect to sensing needle dislodgement, shall be defined as a mechanical response or reaction to a needle, or device of the present disclosure in association with a needle, being dislodged or separated from the tissue of a patient.
[0029] In normal and successful vascular access, the needle delivering fluid into the body is taped to rest flat on the skin surface just behind the access point. Generally, any needle or associated tubing that is not making immediate flat contact with the skin is in danger of being dislodged.
[0030] Because the dislodged needle is not in the body, this fluid may not be reaching its intended destination. If this fluid is blood, this is a highly dangerous condition and should be treated immediately by stopping the fluid flow and re-inserting or replacing the needle.
[0031] The systems of the present disclosure use a skin contact sensor to detect when a vascular access needle is no longer in contact with the skin. In some embodiments, activation of the contact sensor also causes a secondary motion that delivers a compression lever directly onto the soft section of tubing within the flow path in the needle body for automatic restriction/reduction of the fluid flow. With the sensing mechanisms of the present disclosure, the device or add-on device can both detect and immediately stop errant fluid flow due to dislodged access needles.
[0032] FIG. **1**A through FIG. **4** show an embodiment of a sensing mechanism/system **10** employing a spring arm and pinching mechanism in accordance with the present disclosure. Standard needle sets may incorporate sensing mechanism **10** to provide vascular access for blood or saline/medication delivery, and can be manufactured to look and feel almost like existing needle sets yet still incorporate the skin contact sensing mechanism **10** within the needle **20** and tubing **16** set. This sensing mechanism **10** comprises an activation mechanism **25** in the form of a contact member or projecting spring-arm **18** or that is located toward the bottom of housing **12**. The spring-arm **18** size can be adjusted before manufacture to create a device that can precisely determine an exact height for which the needle body **20** has lifted off the surface **50** of the skin during needle dislodgement. For purposes of this disclosure, the term “dislodgement” shall mean a condition where the needle tip has exited the skin, or the housing has lifted off a certain distance from the skin.
[0033] The spring-arm **18** is spring-loaded (e.g. with torsion spring **28** shown in FIG. **3**A and FIG. **4**), and only remains in close apposition to the bottom of the needle body **20** if the tape **52** used to secure the needle **20** it can hold the needle **20** flat down against the skin **50** (see FIG. **1**A). The tape **52** can secure the housing **12** and flaps **14**, holding the housing **12** and flaps **14** against the skin **50** and/or spring arm **18** (see FIG. **1**A′). Housing **12** may also include a cover **32***h*, needle guide **44**, tubing port **42**, and flaps **14**.
[0034] Referring to the dislodged/released orientation shown in FIG. **1**B, when the needle **20** lifts from the body **50**, as occurs during most needle dislodgements, the spring-arm **18** swings open.
[0035] As seen in the sectional views of FIG. **2**A and FIG. **2**B, the spring-arm **18** is coupled mechanically within the interior of housing **12** to a flow termination mechanism comprising a compression lever **24** that forms a pinch valve with tube **40** to stop or significantly limit fluid flow via mechanical disruption of the flow path, e.g. compression or pinching of a soft section **40** of tubing within the needle **20** and flow path. A torsion spring **28** translates movement of the spring-arm **18** into movement of the compression lever **18** via hinge **46** and pins **26** that are disposed in the housing **12** (see FIG. **4**).
[0036] The embodiment shown in FIG. **3**A and FIG. **4** shows compressible tube **40** as a separate structure that is installed into the housing **12**. However, it is contemplated that the housing **12** and tube **40** may be formed from one integral, contiguous piece, by forming a tube **40** from the housing

**12** using a thin-wall tube **40**, thus making it compliant. Ideally, the compression lever **24** would be disposed underneath tube **40** and rotate upward to form a pinch valve, as shown in FIG. **3**B. A two-stage molding process using soft materials for integrated tube section **40** may also be used.

[0037] When tape **52** holds the needle down in the lodged configuration of FIG. **1**A and FIG. **2**A, the spring arm **18** is pushed closed against the housing **12**, thus keeping the compression lever **24** off of tubing **40**. For purposes of clarity, the tape **52** is shown disposed over needle **20** in FIG. **1**A. However, it is appreciated that tape **52** will often be disclosed over the housing **12** and tabs **14**.

[0038] When the tape **52** falls off, and the needle tip **22** is dislodged (configuration of FIG. **1**B and FIG. **2**B), the needle **20** lifts up, allowing the swing arm **18** to open under the pressure of the torsion spring **28**, thus forcing the compression lever **24** downward, pinching tubing **40** and stopping flow to the needle **20**.

[0039] When tape **52** holds the housing **12** down in the lodged configuration of FIG. **1**A′ and FIG. **2**A′, the spring arm **18** is pushed closed against the housing **12**, thus keeping the compression lever **24** off of tubing **40**. For purposes of clarity, the tape **52** is shown disposed over housing **12** in FIG. **1**A′.

[0040] When the tape **52** falls off or is removed from the housing **12** and tabs **14** (or the needle tip **22**, as shown in FIGS. **1**A and **1**B), and the needle tip **22** is dislodged (configuration of FIG. **1**B′ and FIG. **2**B), the housing **12** lifts up, allowing the swing arm **18** to open under the pressure of the torsion spring **28**, thus forcing the compression lever **24** downward, pinching tubing **40** and stopping flow to the needle **20**.

[0041] The integrated design of the compression lever **24** and the spring-arm **18** work synergistically in both detecting problematic dislodgement and immediately solving it. Importantly, system **10** is configured to continuously occlude the tubing at all times during dislodgement. This stops flow and importantly, raises back pressure high enough that a machine pumping the alarm will be triggered to automatically stop pumping.

[0042] FIG. **5**A through FIG. **5**C show an embodiment incorporating a rotary-valve sensing mechanism/system **150**, which includes a housing **152** for receiving tube **16** and needle **20**. Housing **152** further includes a central channel **156** in communication with needle **20** and tube **16** for allowing delivery of fluid there between. An aperture **154** runs through channel **156** in approximately an orthogonal orientation, with the aperture **154** configured to receive rod **168** to form a rotary valve, that is coupled to ratchet wheel **164**. Rod **168** comprises a through hole **162** that has a diameter approximately the size of channel **156**, and is located on rod **168** to line up with channel **156** when in an open configuration, as shown in the side view of FIG. **5**B. A sealing mechanism, such as o-ring **312** shown in FIG. **7**C, may also be disposed on rod **168** to seal the rotary valve aperture **154**.

[0043] One end of rod **168** comprises a ratchet wheel **164**, which along with contact member/button **170** form an activation mechanism to open and close the rotary valve rod **168**. Contact member **170** comprises a compliant, dome-shaped diaphragm that contacts the patient's skin and acts as a return spring that is loaded when pressing inward on the patient. The spring-loaded actuation mechanism of button **170** is further detailed with reference to button **210** shown in FIG. **6**A through **6**C, which operates in a similar, if not identical, manner. Button **170** comprises an elongate catch **172** that emanates from the bottom of the inside wall of the button. Catch **172** has a hooked distal end **174** that is configured to interface with teeth **166** of ratchet wheel **164** of rod **168** to form a rotary valve-based flow termination mechanism.

[0044] When in the compressed state, the catch **162** of button **170** is in an extended linear position toward housing **152** and above ratchet wheel **164**. This forms the open state of the rotary valve as seen in the cut-out side view of FIG. **5**B (ratchet wheel **164** and button **170** removed for clarity), with through hole **162** lined up to be substantially concentric with the central channel **156** of the housing. Fluid flow is thus allowed between the tube **16** and the needle **22**.

[0045] When needle **20** is dislodged from the patient's skin, button **170** expands downward from the housing body **152** to its biased, uncompressed state, which causes the catch **172** to move downward such that distal end **174** catches teeth **166** of ratchet wheel **174** to rotate rod **162** (e.g. 90°) within aperture **154**. This results in the through hole **162** no longer being in alignment with central channel **156**, thus terminating or restricting flow through channel **156** from tube **16** to needle **20** as seen in the cut-out side view of FIG. **5**C (ratchet wheel **164** and button **170** removed for clarity). To reset the flow to open upon re-insertion of the needle **20** and compression of the button **170**, the ratchet wheel **164** may simply be rotated e.g. 90° clockwise to the original position, such that indicia **167** lines up in the proper position. The rotary valve mechanism **150** thus converts the linear motion of button **170** to a rotational motion.

[0046] FIG. **6**A through FIG. **6**C show sectional side views of a slider valve, or shuttle valve-based sensing system **200** in accordance with the present disclosure. FIG. **6**A shows the system **200** in an open configuration, allowing fluid flow between needle **20** and tube **16**. Housing **216** includes a central channel **208** in communication with needle **20** and tube **16**. An aperture **214** runs through channel **208** in approximately an orthogonal orientation, with the aperture **214** configured to receive slider pin **202**. Slider pin **202** comprises a through hole **206** that has a diameter approximately the size of channel **208**, and is located to line up with channel **208** in the open configuration of FIG. **6**A. In this configuration, contact member or button **210**, which is coupled to one end of the slider pin **202**, is compressed against the patient's skin **212**. Button **210** comprises a compliant, dome-shaped wall that acts as a return spring that is loaded when pressing inward on the patient. Slider pin **202** may be molded with button **210** as one contiguous piece of material (e.g. silicone), or may be attached to the button **210** via adhesive or other attachment means.

[0047] As seen in FIG. **6**B, which shows the system **200** in a closed configuration, and FIG. **6**C, which shows the system **200** in an open (dashed line) configuration, the slider pin **202** is configured to reciprocate within aperture **214** of the housing **216**. When needle **20** is dislodged from the patient's skin **212**, button **210** expands, which causes the slider pin **202** to retract out of aperture **214**. This results in the through-hole **206** no longer being in alignment with central channel **208**, thus terminating or restricting flow through channel **208** from tube **16** to needle **20** to form a shuttle valve/flow termination mechanism.

[0048] FIG. **7**A shows an exploded perspective view of a preferred embodiment incorporating a rotary valve-sensing mechanism/system **300**, which includes a housing **304** for receiving tube **16** and needle **20**. Housing **304** further includes tabs **316** and a central channel **308** (FIG. **7**B) in communication with needle **20** and tube **16** for allowing delivery of fluid there between. An aperture **314** runs through channel **308** in approximately an orthogonal orientation, with the aperture **314** configured to receive rod **302** that is coupled to spring-arm/lever **310**. Rod **302** comprises a through hole **306** that has a diameter approximately the size of channel **308**, and is located on rod **302** to line up with channel **308** when in an open configuration, i.e. rod **302** and spring-arm **310**.

[0049] As seen in FIG. **7**B, which shows the contact member or spring-arm **310** of system **300** in a closed configuration (dashed line open away from housing **304**) and open configuration (solid line against the housing **304**), the rod **302** is configured to rotate within aperture **314** of the housing **304**. When needle **20** is dislodged from the patient's skin, lever **310** retracts outward from the housing body (e.g. from a biasing member (not shown) such as a torsion spring or the like), which causes the rod **302** to rotate within aperture **214**. The activation mechanism of spring-arm **310** thus results in the through hole **306** no longer being in alignment with central channel **308**, thus terminating or restricting flow through channel **308** from tube **16** to needle **20**, i.e. forming a flow termination mechanism.

[0050] As shown in FIG. **7**C, the rod **302** may comprise a pair of o-rings **312** that are disposed on the rod **302** at the periphery of the openings of aperture **304** to seal aperture **304** from possible leakage of fluids within the central channel **308**.

[0051] From the description herein, it will be appreciated that that the present disclosure

encompasses multiple embodiments which include, but are not limited to, the following:
[0052] 1. An apparatus for automatic termination of flow for fluid delivery, the apparatus comprising: a housing configured for coupling a fluid delivery tube to a needle configured for subcutaneous delivery of fluid within a tissue of a patient; a spring-loaded activation mechanism coupled to the housing; wherein the activation mechanism comprises a first orientation corresponding to a condition where the housing is disposed substantially adjacent to the tissue and the needle lodged within the tissue; wherein the activation mechanism comprises a second orientation corresponding to a condition where the housing is disposed away from the tissue or the needle is dislodged from the tissue; a flow termination mechanism coupled to the activation mechanism; wherein the flow termination mechanism comprises an open configuration allowing flow from the fluid delivery tube to the needle when the activation mechanism is in the first orientation; and wherein the flow termination mechanism comprises a closed configuration substantially terminating flow from the fluid delivery tube to the needle when the activation mechanism is in the second orientation.
[0053] 2. The apparatus of any preceding embodiment: wherein the activation mechanism comprises a contact member configured to be disposed adjacent the patient's skin when the activation mechanism is in the first orientation; and wherein the contact member articulates with respect to the housing to the second orientation.
[0054] 3. The apparatus of any preceding embodiment: wherein the flow termination mechanism comprises a pinch valve that substantially terminates flow from the fluid delivery tube to the needle when the activation mechanism is in the second orientation.
[0055] 4. The apparatus of any preceding embodiment: wherein the contact member and flow termination mechanism comprise a spring-arm and compression lever to form a pinch valve; wherein the spring-arm is disposed adjacent the housing when positioned in the first orientation; wherein the spring-arm articulates away from the housing in the second orientation; and wherein the compression lever articulates in response to articulation of the swing arm in the second orientation to pinch-off flow from the fluid delivery tube to the needle when the activation mechanism is in the second orientation.
[0056] 5. The apparatus of any preceding embodiment: wherein the housing comprises a compliant tube coupling the fluid delivery tube to the needle; and wherein the compression lever articulates against the compliant tube in the second orientation to terminate flow from the fluid delivery tube to the needle.
[0057] 6. The apparatus of any preceding embodiment: wherein the contact member comprises a dome-shaped button that is biased in an expanded configuration corresponding to the second orientation; and wherein the button is loaded in a compressed configuration adjacent the patient's skin in the first orientation.
[0058] 7. The apparatus of any preceding embodiment: wherein the flow termination mechanism comprises a shuttle valve coupled to the contact member; and wherein the contact member affects translation of the shuttle valve from within the housing from the first orientation to the second orientation.
[0059] 8. The apparatus of any preceding embodiment: wherein the contact member comprises a dome-shaped button that is biased in an expanded configuration corresponding to the second orientation; and wherein the button is loaded in a compressed configuration adjacent the patient's skin in the first orientation.
[0060] 9. The apparatus of any preceding embodiment: wherein the flow termination mechanism comprises a rotary valve coupled the contact member; and wherein the contact member affects rotation of the rotary valve from within the housing from the first orientation to the second orientation.
[0061] 10. The apparatus of any preceding embodiment: wherein the contact member and rotary valve comprise a lever and a rod, the rod being disposed in an aperture within the housing; the

housing comprising a central channel allowing fluid flow from the fluid delivery tube to the needle; wherein the rod comprises a through-hole that is in alignment with the central channel when in the first orientation; wherein the spring-arm is disposed adjacent the housing when positioned in the first orientation and articulates away from the housing in the second orientation; and wherein the rod rotates in response to articulation of the swing arm in the second orientation to rotate the through-hole out of alignment with the central channel to inhibit fluid flow from the fluid delivery tube to the needle when the activation mechanism is in the second orientation.

[0062] 11. The apparatus of any preceding embodiment: wherein the contact member and rotary valve comprise a button and a rod, the rod being disposed in an aperture within the housing; the housing comprising a central channel allowing fluid flow from the fluid delivery tube to the needle; wherein the rod comprises a through-hole that is in alignment with the central channel when in the first orientation; wherein the button is adjacent the housing when positioned in the first orientation and retracts away from the housing in the second orientation; and wherein the rod rotates in response to retraction of the button in the second orientation to rotate the through-hole out of alignment with the central channel to inhibit fluid flow from the fluid delivery tube to the needle when the activation mechanism is in the second orientation.

[0063] 12. A system for automatic termination of flow for a fluid delivery, the system comprising: a fluid delivery tube and a needle configured for subcutaneous delivery of fluid within a tissue of a patient; a housing configured for coupling the fluid delivery tube to the needle; a spring-loaded activation mechanism coupled to the housing; wherein the activation mechanism comprises a first orientation corresponding to a condition where the housing is disposed substantially adjacent to the tissue and the needle lodged within the tissue; wherein the activation mechanism comprises a second orientation corresponding to a condition where the housing is disposed away from the tissue or the needle being dislodged from the tissue; a flow termination mechanism coupled to the activation mechanism; wherein the flow termination mechanism comprises an open configuration allowing flow from the fluid delivery tube to the needle when the activation mechanism is in the first orientation; and wherein the flow termination mechanism comprises a closed configuration substantially terminating flow from the fluid delivery tube to the needle when the activation mechanism is in the second orientation.

[0064] 13. The system of any preceding embodiment: wherein the activation mechanism comprises a contact member configured to be disposed adjacent the patient's skin when the activation mechanism is in the first orientation; and wherein the contact member articulates with respect to the housing to the second orientation.

[0065] 14. The system of any preceding embodiment: wherein the flow termination mechanism comprises a pinch valve that substantially terminates flow from the fluid delivery tube to the needle when the activation mechanism is in the second orientation.

[0066] 15. The system of any preceding embodiment: wherein the contact member and flow termination mechanism comprise a spring-arm and compression lever to form a pinch valve; wherein the spring-arm is disposed adjacent the housing when positioned in the first orientation; wherein the spring-arm articulates away from the housing in the second orientation; and wherein the compression lever articulates in response to articulation of the swing arm in the second orientation to pinch-off flow from the fluid delivery tube to the needle when the activation mechanism is in the second orientation.

[0067] 16. The system of any preceding embodiment: wherein the housing comprises a compliant tube coupling the fluid delivery tube to the needle; and wherein the compression lever articulates against the compliant tube in the second orientation to terminate flow from the fluid delivery tube to the needle.

[0068] 17. The system of any preceding embodiment: wherein the contact member comprises dome-shaped button that is biased in an expanded configuration corresponding to the second orientation; and wherein the button is loaded in a compressed configuration adjacent the patient's skin in the first orientation.

[0069] 18. The system of any preceding embodiment: wherein the flow termination mechanism comprises a shuttle valve coupled to the contact member; and wherein the contact member affects translation of the shuttle valve from within the housing from the first orientation to the second orientation.

[0070] 19. The system of any preceding embodiment: wherein the contact member comprises a spring-loaded, dome-shaped button that is biased in an expanded configuration corresponding to the second orientation; and wherein the button is loaded in a compressed configuration adjacent the patient's skin in the first orientation.

[0071] 20. The system of any preceding embodiment: wherein the flow termination mechanism comprises a rotary valve coupled to the contact member; and wherein the contact member affects rotation of the rotary valve from within the housing from the first orientation to the second orientation.

[0072] 21. The system of any preceding embodiment: wherein the contact member and rotary valve comprise a lever and a rod, the rod being disposed in an aperture within the housing; the housing comprising a central channel allowing fluid flow from the fluid delivery tube to the needle; wherein the rod comprises a through-hole that is in alignment with the central channel when in the first orientation; wherein the spring-arm is disposed adjacent the housing when positioned in the first orientation and articulates away from the housing in the second orientation; and wherein the rod rotates in response to articulation of the swing arm in the second orientation to rotate the through-hole out of alignment with the central channel to inhibit fluid flow from the fluid delivery tube to the needle when the activation mechanism is in the second orientation.

[0073] 22. The system of any preceding embodiment: wherein the contact member and rotary valve comprise a button and a rod, the rod being disposed in an aperture within the housing; the housing comprising a central channel allowing fluid flow from the fluid delivery tube to the needle; wherein the rod comprises a through-hole that is in alignment with the central channel when in the first orientation; wherein the button is adjacent the housing when positioned in the first orientation and retracts away from the housing in the second orientation; and wherein the rod rotates in response to retraction of the button in the second orientation to rotate the through-hole out of alignment with the central channel to inhibit fluid flow from the fluid delivery tube to the needle when the activation mechanism is in the second orientation.

[0074] 23. A method for automatic termination of flow for fluid delivery within a patient, the method comprising: coupling a housing to a surface of a patient's tissue; the housing configured for coupling a fluid delivery tube to a needle for delivery of fluid within the tissue of a patient; preloading a spring-loaded activation mechanism at a first orientation corresponding to a condition where the housing is disposed adjacent to the tissue and the needle lodged within the tissue; wherein a flow termination mechanism coupled to the activation mechanism is disposed in an open configuration allowing flow from the fluid delivery tube to the needle when the activation mechanism is in the first orientation; upon release of the housing away from the tissue or the needle being dislodged from the tissue, advancing the activation mechanism to a second orientation; and switching the flow termination mechanism to a closed configuration to substantially terminate flow from the fluid delivery tube to the needle.

[0075] 24. The method of any preceding embodiment: wherein the flow termination mechanism comprises a pinch valve; and wherein switching the flow termination mechanism to a closed configuration comprises pinching-off flow between the fluid delivery tube and the needle.

[0076] 25. The method of any preceding embodiment: wherein the activation mechanism comprises a contact member disposed adjacent the housing when positioned in the first orientation; wherein the contact member articulates away from the housing in the second orientation; wherein the housing comprises a compliant tube coupling the fluid delivery tube to the needle; and wherein flow termination mechanism comprises a compression lever that articulates against the compliant

tube in the second orientation to terminate flow from the fluid delivery tube to the needle.
[0077] 26. The method of any preceding embodiment: wherein the activation mechanism comprises a contact member disposed adjacent the housing when positioned in the first orientation; wherein the contact member articulates away from the housing in the second orientation; wherein the flow termination mechanism comprises a rotary valve coupled to the contact member; and wherein the contact member affects rotation of the rotary valve from within the housing from the first orientation to the second orientation.
[0078] 27. The method of any preceding embodiment: wherein the contact member and rotary valve comprise a lever and a rod, the rod being disposed in an aperture within the housing; the housing comprising a central channel allowing fluid flow from the fluid delivery tube to the needle; wherein the rod comprises a through-hole that is in alignment with the central channel when in the first orientation; wherein the spring-arm is disposed adjacent the housing when positioned in the first orientation and articulates away from the housing in the second orientation; and wherein the rod rotates in response to articulation of the swing arm in the second orientation to rotate the through-hole out of alignment with the central channel to inhibit fluid flow from the fluid delivery tube to the needle when the activation mechanism is in the second orientation.
[0079] Although the description herein contains many details, these should not be construed as limiting the scope of the disclosure but as merely providing illustrations of some of the presently preferred embodiments. Therefore, it will be appreciated that the scope of the disclosure fully encompasses other embodiments which may become obvious to those skilled in the art.
[0080] In the claims, reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” All structural, chemical, and functional equivalents to the elements of the disclosed embodiments that are known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the present claims. Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed as a “means plus function” element unless the element is expressly recited using the phrase “means for”. No claim element herein is to be construed as a “step plus function” element unless the element is expressly recited using the phrase “step for”.

## Claims

**1**. A system, device, and/or method disclosed herein.